Old-Age, Widow and Disability Pensions under India's National Social Assistance Programme (NSAP): Is Coverage Reaching Those in Need?

A State-Level Exploratory Study Using Data Analysis and K-Means Clustering

ARUNIMA R

Department of Computer Science, Ramanujan College, University of Delhi, New Delhi, India

Roll No.: 24570010

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Abstract

The National Social Assistance Programme (NSAP) is an important social protection initiative in India that provides financial assistance to elderly people, widows and persons with disabilities. This study examines the state-wise distribution of beneficiaries under three major NSAP pension schemes: IGNOAPS, IGNWPS and IGNDPS. It uses secondary data from government sources and the National Multidimensional Poverty Index (MPI) to explore variations in pension beneficiary distribution across Indian States and Union Territories. Python-based data analysis, data visualisation and K-means clustering are used to identify patterns and similarities among states. The study aims to understand the relationship between pension distribution and multidimensional poverty and highlight potential gaps in social protection. The findings provide an exploratory perspective on the distribution of welfare benefits and the need for more inclusive social assistance.

Keywords: National Social Assistance Programme, NSAP, Old-age Pension, Widow Pension, Disability Pension, Social Protection, Multidimensional Poverty, K-means Clustering, Machine Learning, India.

1. Introduction

Poverty and economic insecurity remain important challenges in developing countries, particularly among elderly people, widows and persons with disabilities. These groups may face difficulties in earning a regular income and accessing financial resources. Social protection programmes are therefore important for providing financial assistance and reducing economic vulnerability.

In India, the National Social Assistance Programme (NSAP) is a centrally sponsored social assistance programme implemented by the Ministry of Rural Development. It aims to provide

financial support to vulnerable sections of society through different schemes.

This study focuses on three major pension schemes under NSAP:

- **Indira Gandhi National Old Age Pension Scheme (IGNOAPS):** Provides financial assistance to eligible elderly people.
- **Indira Gandhi National Widow Pension Scheme (IGNWPS):** Provides financial assistance to eligible widows.
- **Indira Gandhi National Disability Pension Scheme (IGNDPS):** Provides financial assistance to eligible persons with disabilities.

Although these schemes provide support to millions of people, the number of beneficiaries differs across states and Union Territories. Such differences may be influenced by population size, poverty levels, eligibility conditions and the implementation of welfare programmes.

By studying beneficiary distribution alongside poverty indicators, it is possible to explore whether states with different socioeconomic conditions also show different patterns of pension assistance.

This research is aligned with Sustainable Development Goal 1 (No Poverty) and Sustainable Development Goal 10 (Reduced Inequalities).

The main objective is to investigate state-wise differences in NSAP pension beneficiaries and explore whether states can be grouped according to similarities in pension distribution and poverty indicators.

2. Related Work

Previous studies have examined the role of social pensions in reducing poverty and supporting vulnerable populations in India. Garroway (2013) evaluated the National Social Assistance Programme using household-level data and found variations in the effectiveness of old-age and widow pensions across income groups, geographical locations, and social categories. The study also highlighted limitations in targeting benefits towards poor households, underscoring the importance of examining the relationship between social assistance and socioeconomic need.

Asher, Vora, and Maurya (2015) discussed pension and healthcare initiatives for informal-sector workers in India, highlighting concerns regarding the limited coverage of existing social protection programmes and the need to strengthen support for vulnerable populations. Their analysis provides a broader context for examining the adequacy of non-contributory pension coverage in an ageing society.

At the national level, NITI Aayog's *National Multidimensional Poverty Index: A Progress Review 2023* provides state- and Union Territory-level estimates of multidimensional poverty based on the National Family Health Survey (NFHS-5), conducted during 2019–2021. These indicators offer a potential basis for comparing regional poverty conditions with the distribution of NSAP beneficiaries, although multidimensional poverty estimates do not directly measure poverty among older adults or determine pension eligibility.

However, poverty indicators, demographic statistics and pension beneficiary counts measure different aspects of society. Their combination must therefore be interpreted carefully. This study uses these sources for exploratory analysis rather than treating beneficiary numbers as direct measurements of pension coverage.

3. Dataset and Preprocessing

Data Sources and Description

The study uses three secondary datasets from government sources. The NSAP beneficiary dataset contains 36 State/Union Territory observations and one national summary row, covering three pension schemes: IGNOAPS (old-age), IGNWPS (widow) and IGNDPS (disability). The national summary row is excluded from the state-level analysis. The second dataset contains state-wise Multidimensional Poverty Index (MPI) headcount ratios based on NFHS-5 data (2019–

21), providing socioeconomic indicators for comparison. The third dataset uses Census of India 2011 age-wise population information as a historical demographic reference to understand the elderly population. Since the Census data is from 2011 and the NSAP reporting period is unverified, it is not used to calculate direct pension coverage rates.

Table 1. Variables used in the study

Variable	Description	Type
State_UT	Name of the state or Union Territory	Categorical
Old-age beneficiaries	IGNOAPS beneficiary count	Numerical
Widow beneficiaries	IGNWPS beneficiary count	Numerical
Disability beneficiaries	IGNDPS beneficiary count	Numerical
MPI headcount ratio	Percentage of population experiencing multidimensional poverty	Numerical
Cluster	Cluster assigned to each state	Categorical

Data Cleaning and Integration

The datasets were prepared before performing the analysis.

The following steps were carried out:

1. **Data collection:** NSAP beneficiary data and MPI poverty indicators were obtained from government sources.
2. **Data selection:** The three major pension schemes were selected for the study.
3. **Removal of summary records:** The national total row was excluded to avoid treating it as an individual state.
4. **Data cleaning:** Unnecessary columns and irrelevant records were excluded.
5. **State name matching:** State and Union Territory names were harmonised to allow datasets to be combined.
6. **Data integration:** NSAP beneficiary counts were combined with MPI indicators using state/Union Territory names.
7. **Missing-value checking:** Missing values were examined and were not automatically treated as zero.
8. **Feature preparation:** The selected numerical variables were prepared for descriptive analysis and clustering.

The final merged dataset contains 36 state/Union Territory records.

Feature Construction

The main features used in the analysis are the number of old-age pension beneficiaries, widow pension beneficiaries, disability pension beneficiaries and the MPI headcount ratio.

A combined pension beneficiary count can be calculated as:

Total Pension Beneficiaries = Old-age Beneficiaries + Widow Beneficiaries + Disability Beneficiaries

This variable provides an overall summary of the three pension schemes.

The MPI headcount ratio is retained as a separate indicator because it represents general multidimensional poverty rather than pension eligibility.

The clustering results are based on the features represented in the supplied clustering output. The exact feature-scaling and parameter settings should be documented from the code used to produce that output.

4. Methodology

The study follows a data-analysis approach involving descriptive statistics, data visualisation and unsupervised machine learning.

Descriptive Statistical Analysis

Descriptive analysis is used to understand the distribution of beneficiaries across states and pension schemes.

The analysis includes:

- Comparing beneficiary counts across the three pension schemes.
- Identifying states with high and low beneficiary numbers.
- Examining differences between states and Union Territories.
- Comparing beneficiary distribution with MPI headcount ratios.

K-means Clustering

K-means is an unsupervised machine-learning algorithm used to divide observations into groups called clusters.

In this study, each state or Union Territory is treated as an observation. Pension beneficiary counts and the MPI headcount ratio are used as numerical characteristics for grouping the states.

The objective is to group states that have similar values in the selected features.

The K-means algorithm works through the following steps:

1. Select the numerical features.
2. Choose the number of clusters, K.
3. Initialise the cluster centroids.
4. Calculate the distance between each state and the cluster centroids.
5. Assign each state to the nearest centroid.
6. Recalculate the centroids using the assigned states.
7. Repeat the process until the cluster assignments stabilise.

The distance between observations is commonly calculated using Euclidean distance:

$$d(x, y) = \sqrt{\Sigma(x_i - y_i)^2}$$

The clustering output assigns each state to one of three groups. These groups are examined to understand similarities in pension beneficiary distribution and poverty indicators.

Regression Analysis: Proposed Extension

Regression methods such as Linear Regression and Ridge Regression were part of the broader project plan. However, a valid regression experiment requires a clearly defined dependent variable and sufficient explanatory variables.

The current merged dataset does not contain scheme-specific eligible-population denominators. Therefore, it cannot directly establish a reliable pension coverage rate.

5. Experimental Setup

The analysis uses a merged state/Union Territory-level dataset containing 36 observations.

The selected numerical variables include:

- Old-age pension beneficiary count.
- Widow pension beneficiary count.
- Disability pension beneficiary count.
- MPI headcount ratio.

Descriptive analysis and visualisation are used to examine state-wise differences. K-means clustering is used to group states according to similarities in the selected variables.

The clustering output contains three cluster labels: 0, 1 and 2.

For a complete experimental record, the following settings should be reported from the original code:

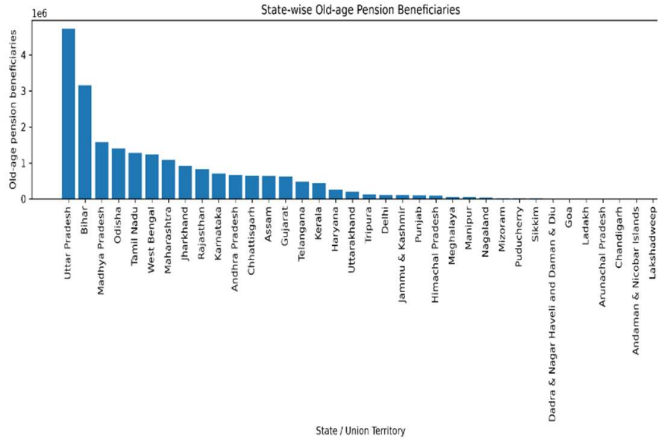
- Number of clusters (K): 3.
- Feature-scaling method: [Confirm from code].
- Distance measure: [Confirm from implementation].
- Random initialisation and random seed: [Confirm from code].

- Silhouette score: [Include only if calculated].

The analysis is exploratory and does not claim predictive accuracy or causal relationships.

6. Results and Analysis

Figure 1. State-wise Old-age Pension Beneficiaries



The graph presents the distribution of old-age pension beneficiaries across Indian states and Union Territories.

Uttar Pradesh records the highest number of old-age pension beneficiaries, with approximately 4.72 million beneficiaries. Bihar ranks second with approximately 3.16 million. Madhya Pradesh and Odisha also show comparatively high beneficiary counts.

The remaining states show different levels of beneficiary distribution, with several smaller states and Union Territories recording substantially lower counts.

The graph indicates that old-age pension beneficiaries are not evenly distributed across states. However, the number of beneficiaries is influenced by population size and the number of eligible elderly people. Therefore, higher beneficiary counts do not necessarily indicate better pension coverage.

Overall Distribution Across Pension Schemes

The supplied NSAP dataset reports the following beneficiary totals:

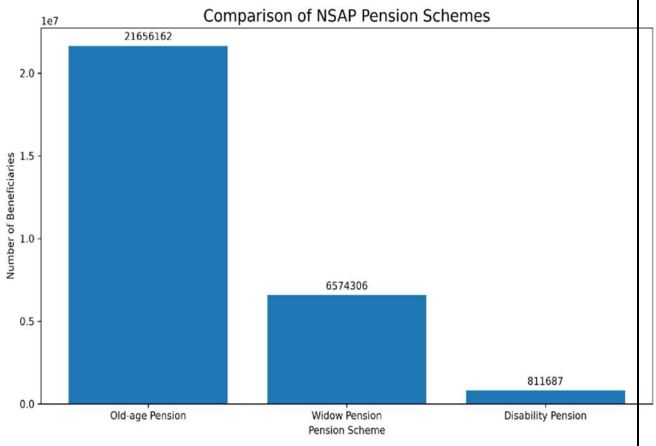
Table 2. Beneficiaries under the three core pension schemes

Pension Scheme	**Reported Beneficiaries**
Old-age pension (IGNOAPS)	21,656,162
Widow pension (IGNWPS)	6,574,306
Disability pension (IGNDPS)	811,687
Combined total	**29,042,155**

The old-age pension scheme has the largest reported beneficiary count, followed by the widow pension scheme. The disability pension scheme has the lowest count among the three schemes.

These totals indicate the relative scale of the three pension categories in the supplied dataset. They should not be interpreted as the proportion of eligible people receiving assistance.

Figure 2. Comparison of Beneficiaries Across the Three NSAP Pension Schemes



Multidimensional Poverty Analysis

The MPI headcount ratio provides information about the level of multidimensional poverty across Indian states.

According to the supplied MPI dataset, some of the states with higher poverty headcount ratios include:

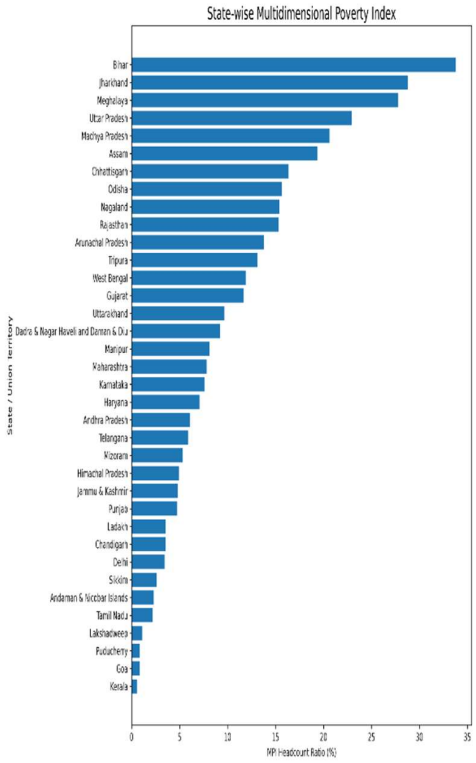
Table 3. Selected states with higher MPI headcount ratios

State	MPI Headcount Ratio
Bihar	33.76%
Jharkhand	28.81%
Uttar Pradesh	22.93%
Madhya Pradesh	20.63%
Assam	19.35%

Bihar records the highest MPI headcount ratio among the states listed, followed by Jharkhand and Uttar Pradesh.

These figures provide socioeconomic context for the pension analysis. However, MPI measures multidimensional poverty in the general population and is not a direct measure of the number of people eligible for NSAP pensions.

Figure 3. State-wise Multidimensional Poverty Headcount Ratio



K-means Clustering Results

K-means clustering was used to group the 36 states and Union Territories according to similarities in the selected numerical variables.

The supplied clustering results contain three groups.

Table 4. State distribution across clusters

Cluster	Number of States/UTs	States/UTs
Cluster 0	11	Assam, Chhattisgarh, Jharkhand, Gujarat, Karnataka, Madhya Pradesh, Meghalaya, Odisha, Rajasthan, Tamil Nadu, West Bengal
Cluster 1	2	Bihar, Uttar Pradesh
Cluster 2	23	Andhra Pradesh, Arunachal Pradesh, Andaman and Nicobar Islands, Chandigarh, Dadra and Nagar Haveli and Daman and Diu, Delhi, Goa, Haryana, Himachal Pradesh, Jammu and Kashmir, Kerala, Ladakh, Lakshadweep, Maharashtra, Manipur, Mizoram, Nagaland, Puducherry, Punjab, Sikkim, Telangana, Tripura, Uttarakhand

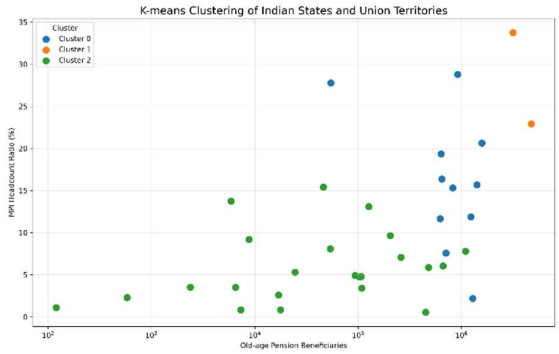
The clustering output shows that Bihar and Uttar Pradesh are placed together in Cluster 1. Both states have comparatively high old-age pension beneficiary counts and high MPI headcount ratios in the supplied dataset.

Cluster 0 contains eleven states with varied beneficiary counts and poverty indicators. These states show similarities according to the numerical features used by the clustering algorithm.

Cluster 2 contains twenty-three states and Union Territories. This group includes several states and territories with comparatively smaller beneficiary counts, although the values differ within the group.

The results demonstrate that clustering can identify groups of states with similar characteristics. However, cluster labels are algorithm-generated identifiers and do not represent official categories of pension coverage.

Figure 4. K-means Clustering of Indian States and Union Territories



Interpretation of Results

The results indicate considerable variation in NSAP pension beneficiary numbers across Indian states.

Uttar Pradesh and Bihar have particularly high old-age pension beneficiary counts. The clustering output also places these two states in the same group, suggesting similarities in the selected numerical indicators.

The poverty indicators provide additional information about socioeconomic conditions. Bihar, Jharkhand and Uttar Pradesh have relatively high MPI headcount ratios in the supplied data.

7. Discussion

The study examines differences in the distribution of old-age, widow and disability pension beneficiaries across Indian states and Union Territories.

The descriptive analysis shows that old-age pensions account for the largest beneficiary group among the three selected schemes. Uttar Pradesh records the highest old-age pension beneficiary count, followed by Bihar, Madhya Pradesh and Odisha.

The clustering results provide another perspective on state-level differences. Bihar and Uttar Pradesh form one cluster, while the other states are distributed across two additional clusters. This suggests that the selected variables can be used to explore similarities between states.

The comparison with MPI indicators provides a broader understanding of socioeconomic conditions. States with high multidimensional poverty may require particular attention when assessing the reach of social protection programmes.

However, the relationship between poverty and beneficiary counts should not be interpreted as a direct relationship between poverty and pension coverage. A large state may have more beneficiaries simply because it has a larger population.

The findings therefore suggest that beneficiary distribution and poverty indicators should be examined together with demographic and eligibility information. This would provide a more meaningful understanding of whether social assistance is reaching the intended population.

The study also demonstrates the usefulness of machine-learning techniques in exploring public welfare datasets. K-means clustering can help identify groups of states for further policy investigation, but it cannot independently determine programme effectiveness.

8. Limitations

The study has several limitations.

1. The NSAP dataset contains aggregate beneficiary counts rather than individual-level information.
2. The exact reference period of the supplied NSAP extract needs to be confirmed from its metadata.
3. The MPI data are based on NFHS-5 for 2019–21 and may not match the NSAP reporting period.

4. MPI represents general multidimensional poverty rather than pension-specific eligibility.
5. Scheme-specific eligible-population denominators are not available in the merged dataset.
6. Population differences across states may affect the number of reported beneficiaries.
7. The clustering output depends on the selected features, scaling method and number of clusters.
8. The present analysis is exploratory and does not establish causal relationships.
9. Regression performance metrics have not been reported because a validated regression experiment is not available in the current results.

These limitations should be considered when interpreting the findings.

9. Conclusion and Future Work

This study analyses the state-wise distribution of beneficiaries under three major pension schemes of India's National Social Assistance Programme: old-age, widow and disability pensions.

The analysis shows that the number of beneficiaries varies considerably across states. Uttar Pradesh and Bihar have the highest old-age pension beneficiary counts in the supplied dataset. The overall scheme comparison also indicates that old-age pensions account for the largest beneficiary group.

The K-means clustering results divide the 36 states and Union Territories into three groups based on similarities in the selected variables. Bihar and Uttar Pradesh form one cluster, while the remaining states are distributed across two other groups.

The comparison with multidimensional poverty indicators provides additional socioeconomic context. However, the available information is not sufficient to determine conclusively whether pensions are reaching all eligible people in need.

Future research should incorporate updated population data, scheme-specific eligible-population estimates and consistent reporting periods. Further analysis could examine gender differences, rural and urban distribution, regional inequalities and changes in beneficiary coverage over time.

The study demonstrates how data analysis and machine-learning techniques can support the exploration of social protection programmes and contribute to the broader goals of poverty reduction and reduced inequalities.

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